M3 Critical Evaluation and Transfer: High-Level Synthesis with Game Theoretic Scheduling

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***Abstract*—[Milestone 3: Critical Evaluation and Transfer. Judges the game theoretic HLS scheduling/binding approach: its strengths, limitations, and transfer to embedded systems. The experimental results section is extended using the benchmark data of Murugavel and Ranganathan [1] (differential equation, FIR, IIR, lattice, EWF, WAVE, NC filter) across scheduling, binding, and combined scheduling+binding. Structured bullets and tables; AI usage in accompanying protocol.]**

I. Concrete Strengths

1) **Joint scheduling and binding.**
   - Single auction framework handles both, unlike ILP/LP that treat them separately [2].
   - Combined formulation applies the Nash equilibrium once and gives better power reduction than running the two in sequence.
2) **Direct power objective.**
   - Each unit bids by actual power cost; the equilibrium minimises total power across all players.
   - Evidence: 6.3% (scheduling), 13.9% (binding), 11.8% (combined) improvement [1].
3) **No area overhead.**
   - No extra modules or multiplexers added; register count unchanged.
   - Multiplexer power varies minimally with binding and is small vs. functional units.
4) **Comparable runtime.**
   - Player count restricted to 2–3 units per type per game $\Rightarrow$ small games $\Rightarrow$ runtimes comparable to ILP/LP.
5) **Power-reduction techniques integrated.**
   - Functional unit sharing, path balancing, and register assignment embedded in the cost function.

II. Concrete Limitations, Risks, and Hidden Assumptions

1) **Scalability of the Nash equilibrium.**
   - Nash equilibrium is NP-complete; complexity $\mathcal{O}(N^S NS)$ per game [1].
   - For large circuits with many players, finding the equilibrium becomes slow.
2) **Latency increase.**
   - Power is optimised, not latency; worst-case increase about 2 cycles.
   - Unsuitable where latency is a hard constraint.
3) **Missing interconnect model.**
   - Wire delay and interconnect power not modelled; significant in deep-submicron / nano regimes.
   - Optimisation is therefore incomplete for modern process nodes.
4) **Resource-constraint assumption.**
   - Assumes only a few modules of each type; if only one multiplier exists the binding game is trivial.
5) **Simulation-dependent costs.**
   - Switching / neighbourhood-input power values come from simulation (100k vectors); accuracy depends on input statistics.

III. Scalability Analysis

- **Time complexity**: scheduling/binding $\mathcal{O}(lcN^S NS)$; combined $\mathcal{O}(lcN^S NS)$, where l = levels, c = compatible-operation sets per level [1].
- **Payoff-matrix construction**: $\mathcal{O}(MO)$ — quadratic in modules M and operations O.
- **Space complexity**: $\mathcal{O}(MO)$ — the size of the payoff matrix per game.
- Practical runtimes stay small because each game restricts players to < 4, with multiple small games per control step.

IV. Experimental Results (Extended)

Benchmarks: differential equation (Diff. eqn), FIR, IIR, lattice, elliptic wave filter (EWF), WAVE, and noise-cancel filter (NC filter). Power obtained by simulation with 100 000 random 16-bit two's-complement vectors using TSMC 0.25 μm cells; interconnect length estimated via Rent's rule [1].

TABLE I
SCHEDULING: POWER AND % IMPROVEMENT OVER ILP [1]

Circuit	Constr. (A,M)	ILP P_{ILP}	GT P_{GT}	Impr.
Diff. eqn	2,2	24.8	23.9	3.6%
FIR	3,3	82.8	78.7	4.9%
IIR	2,2	15.6	14.8	5.1%
Lattice	3,3	91.3	82.9	9.2%
EWF	2,2	262.8	251.0	4.5%
WAVE	2,2	258.2	232.8	9.8%
NC filter	2,2	389.7	363.4	6.7%
			Average	**6.3%**

A. Results for Scheduling (vs. ILP)

- Power (mW); A = adders, M = multipliers in the resource constraint.
- Average 6.3% improvement; latency occasionally slightly higher than ILP.
- Runtimes for ILP and GT are comparable.

B. Results for Binding (vs. LP)

TABLE II
BINDING: POWER AND % IMPROVEMENT OVER LP [1]

Circuit	LP P_{LP}	GT P_{GT}	Impr.
Diff. eqn	22.9	20.1	12.2%
FIR	72.0	57.7	19.8%
IIR	15.5	12.3	20.6%
Lattice	71.9	64.1	10.8%
EWF	234.3	210.0	10.4%
WAVE	215.1	196.3	8.7%
NC filter	368.5	312.8	15.1%
		Average	**13.9%**

- Power (mW); compared against random, greedy, and LP-based binding; LP shown as the strongest baseline.
- Average 13.9% improvement, even without interconnect power optimisation.
- Gain attributed to the Nash equilibrium optimising for all players plus joint functional-unit and register binding.

C. Results for Combined Scheduling + Binding (vs. ILP)

TABLE III
COMBINED: POWER AND % IMPROVEMENT OVER ILP [1]

Circuit	Constr.	ILP P_{ILP}	GT P_{GT}	Impr.
Diff. eqn	2,2	22.9	16.0	8.0%
FIR	3,3	61.7	50.4	18.3%
IIR	2,2	12.8	11.2	12.5%
Lattice	3,3	66.3	55.9	15.7%
EWF	2,2	204.0	191.3	6.2%
WAVE	2,2	201.2	179.2	10.9%
NC filter	2,2	324.3	286.7	11.5%
			Average	**11.8%**

- Power (mW); ILP baseline = ILP scheduling + LP binding in sequence.
- Average 11.8% improvement; worst-case latency increase ≤ 2 cycles.
- Nash equilibrium computed via the Gambit game-theory toolkit [3].

D. Result Summary

TABLE IV
AVERAGE POWER IMPROVEMENT OF THE GAME-THEORETIC APPROACH

Task	Average improvement
Scheduling (vs. ILP)	6.3%
Binding (vs. LP)	13.9%
Combined (vs. ILP)	11.8%

- No area increase across all benchmarks; runtimes comparable to ILP/LP.
- Only cost: small latency increase (worst case $\sim$2 cycles).
- Largest single-circuit binding gain: IIR at 20.6%.

V. APPLICATION SUITABILITY

A. Suitable

1) **Battery-powered embedded devices**: IoT sensors, smartwatches, mobile chips — small batteries, strict power limits.
2) **Cost-sensitive ASICs**: power reduction with no area increase lowers per-unit and cooling cost.
3) **DSP datapath kernels**: FIR/IIR filters, transforms — the benchmark profile.
4) **AI accelerators**: power-aware datapath scheduling for edge inference.

B. Unsuitable / Needs Adaptation

1) **Latency-critical hard-real-time designs**: power-first objective trades away latency.
2) **Very large circuits**: NP-complete Nash equilibrium becomes slow without faster computation.
3) **Deep-submicron interconnect-dominated designs**: needs interconnect binding (not yet modelled).
4) **Single-unit-per-type architectures**: binding game becomes trivial.

VI. EXTENSIONS AND OPEN QUESTIONS

1) **Joint power + latency**: reformulate for simultaneous optimisation of both.
2) **Interconnect binding**: add wire delay and interconnect power to the cost function.
3) **Faster equilibrium for large circuits**: approximate or decomposed Nash computation.
4) **Machine-learning-guided bidding**: learn from past designs to make smarter scheduling/binding decisions.

TABLE V
M3 EVALUATION SUMMARY

Dimension	Assessment
Core idea	Sound: auction + Nash equilibrium for power-aware scheduling/binding
Power gains	6.3% / 13.9% / 11.8% (sched / bind / combined)
Area	No overhead
Runtime	Comparable to ILP/LP for small player counts
Latency	Slight increase (≤ 2 cycles)
Scalability	Limited by NP-complete Nash equilibrium
Interconnect	Not modelled — key gap
Extension potential	High: latency co-opt., interconnect, ML bidding

VII. EVALUATION SUMMARY

- **Overall**: a useful, power-effective approach to HLS scheduling and binding with no area cost.
- Main weaknesses are scalability (Nash equilibrium complexity) and the missing interconnect model.
- Best understood as an effective low-power method for small-to-medium DSP datapaths.

REFERENCES

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